

Issues identified & Fixes implemented: Task management application.

Variables & Operators

Errors found

- taskList declared with no var/let/const (implicit global); var used for taskCounter.
- == used instead of === in findTaskByTitle and isHighPriority (utils.js).
- Assignment operator (=) used inside a conditional in updateTaskPriority instead of ===.
- No typeof / parameter validation anywhere in the starter code.

Fixes implemented

- Replaced every var with const or let; removed the implicit global by declaring taskList properly inside its module.
- Replaced == with === throughout (findTaskByTitle, isHighPriority.).
- Replaced the assignment with a strict-equality comparison (task.id === taskId) in updateTaskPriority.
- Added typeof/Array.isArray/instanceof checks (5+) to exported functions, throwing descriptive TypeErrors on bad input.

Control flow

Errors found

- displayAllTasks looped with i <= taskList.length (off-by-one, reads past array end).
- findTaskByTitle's while loop never incremented i (infinite loop).
- calculateAveragePriority divided by taskList.length with no empty-array guard.

Fixes implemented

- Rewrote displayAllTasks as a for...of loop, eliminating the off-by-one entirely.
- Restored the missing i++ and switched the loop's comparison to ===, so it now terminates correctly.
- Added an empty-array check that returns 0, and rounds the result with Math.round instead of returning a raw float.

Functions

Errors found

- findTaskByTitle missing its title parameter entirely.
- countCompletedTasks recursion had no base case and no null/undefined check.
- getHighPriorityTasks and mergeTasks used manual index loops instead of filter/spread.

- No pure functions, higher-order functions, or array methods (map/filter/reduce/find/some/every) anywhere.

Fixes implemented

- Added the title parameter and validated that it is a string before use.
- Added a base case (index >= tasks.length) and a guard for missing array entries, so recursion now terminates safely.
- getHighPriorityTasks now uses Array.filter; mergeTasks(...taskLists) uses rest parameters plus the spread operator via reduce.
- Added 3+ pure functions (getTaskDetails, calculateAveragePriority, mergeTasks) and a higher-order function, createPriorityComparator, that returns a sort comparator; TaskManager now uses map/filter/reduce/find/some/every.

OOP & Classes

Errors found

- Task class missing an id property and a toggleCompletion method.
- SubTask extends Task but never calls super(), which throws in a derived class.
- The TaskManager object implemented only one method.

Fixes implemented

- Added an id (via a new generateRandomId helper), a createdAt timestamp, and a toggleCompletion() method to Task.
- Added the required super(title, description, priority) call in SubTask's constructor, and overrode getInfo() to demonstrate method overriding via super.getInfo().
- Added addTask, getCompletedTasks, getTaskTitles, getTotalPriorityScore, findById, hasHighPriorityTasks, areAllTasksCompleted, and getStatistics to TaskManager (8 methods total).

Modern JavaScript

Errors found

- getInfo() used string concatenation instead of a template literal.
- No destructuring, spread, or rest operators used anywhere.
- No ES6 module import/export statements in any file.

Fixes implemented

- getInfo() and every dynamically built string in dom.js now use template literals (5+ uses total).
- Added object destructuring in getTaskDetails and dom.js handlers, the spread operator in mergeTasks and TaskManager, and a rest parameter in mergeTasks(...taskLists).
- Converted app.js, dom.js, utils.js, and the new storage.js into ES6 modules with proper import/export statements (4 modules total).

DOM & Storage

Errors found

- `getElementById(".add-task-btn")` mixed an ID lookup with a class selector; `querySelector("task-input")` was missing its "#".
- No null checks before `addEventListener` or DOM access.
- Only one event listener existed; no form submit, delete, or filter handling; no event delegation.
- `displayTasks` never cleared existing content before re-rendering.
- Script ran `setupEventListeners()` at parse time instead of on `DOMContentLoaded`.
- `saveToStorage/loadFromStorage` skipped `JSON.stringify/JSON.parse` entirely.
- `generateRandomId` returned a raw `Math.random()` decimal, not a usable id.
- `formatTaskName` was a no-op that returned its input unchanged.

Fixes implemented

- Corrected both selectors (`querySelector(".add-task-btn")`, `getElementById` with the matching id) throughout `dom.js`.
- Added null checks (2+) before every `addEventListener` call and before reading/writing DOM elements.
- Added 5+ listeners: add button, form submit, one delegated click listener on the task list, each filter button, and a storage-sync listener.
- `displayTasks` now clears `container.innerHTML` before rendering, using template literals + `insertAdjacentHTML`.
- `handleTaskListClick` uses `event.target.closest('[data-action]')` for real event delegation on complete/delete actions.
- Initialization now runs inside `document.addEventListener("DOMContentLoaded", initializeApp)`.
- `saveToStorage/loadFromStorage` now use `JSON.stringify/JSON.parse` wrapped in `try/catch`, added `tasksToJSON/tasksFromJSON`, and a new `storage.js` module persists and rehydrates real `Task` instances.
- `generateRandomId` now returns a unique integer from `Date.now()` plus a random offset.
- `formatTaskName` now trims whitespace and capitalizes each word.

HTML & CSS

Errors found

- No priority input field or statistics container in `index.html`.
- Two classic `<script>` tags loaded in a fragile, dependency-unaware order.
- `styles.css` targeted `#app`, which does not exist in the markup (`.container` does).
- Fixed-width layout only, with no responsive breakpoints or design tokens.

Fixes implemented

- Added a priority <select>, a #form-error element, and a #statistics container to index.html.
- Replaced both scripts with a single <script type="module" src="src/dom.js">; the browser resolves the import graph itself.
- Rebuilt styles.css around the actual .container markup and a CSS custom-property token system (color, spacing, radius, shadow).
- Made the layout mobile-first with breakpoints at 640px and 960px, added :focus-visible outlines and prefers-reduced-motion support.

Testing

Errors found

- app.test.js had no imports of the code under test.
- No beforeEach to reset shared state between tests.
- Almost no real assertions; most functionality (classes, inheritance, array operations, recursion, destructuring/spread, error handling) was untested.

Fixes implemented

- Added proper imports from src/app.js and src/utls.js.
- Added beforeEach blocks that rebuild a fresh task list / TaskManager state before every test.
- Wrote 33 tests total, including 3+ edge-case tests, covering classes, SubTask inheritance, array operations, the recursive function, destructuring/spread, and error handling — all passing.